

Matt Owen

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tl;dr

ML Engineering Manager and AI Hacker, solving the hardest problems by delivering bleeding-edge research into production. I have worked for 20+ years as a Researcher, Engineer, and Manager, tackling problems at every level: real-time scale, frontier innovation, and shipping usable products. My teams and I have consistently solved complex problems, owned a ton of infrastructure, and celebrated the delivery of ambitious projects. These days I'm bringing closed-source-level performance to on-prem: fine-tuned open-weight models and evaluation stacks that run entirely inside the boundary, no data egress. Whatever my ikigai is, it braids Mathematics and Programming; lfg.

ML + AI	LLMs, Generative AI , AI Agents, RAG, MCP, Fine-tuning, PyTorch, vLLM, Evals, Knowledge Graphs
Engineering + Platform	Terraform , Kubernetes , MLOps, Airflow, Kafka , Celery , AWS, Spark, Golang , Python
Achievements	• Building Clearwing, an agentic SOC and pentester • Curated data collection and evaluation of unaligned models - covered in The Economist • Fine-tuned Qwen3 for vulnerability scanning and tool usage • Built company-wide data pipeline and lake, unifying Research, Product, and Engineering • Architected and deployed massive-scale, real-time data pipelines: 1.2M+ records/second • Launched innovative, agent-based, llm-powered products • Maintained infrastructure comprising 1,000+ instances • Freelanced as a Founding AI Engineer, building ML applications for seed-stage companies • Fine-tuned Mixtral forever-ago for use as a POC red-team model
Education	Bachelor of Science, Mathematics - Boston University

Work Experience

Lazarus AI ML Engineering Manager October 2025 - Present

[Leading ML engineering and applied research](#) across a broad portfolio: Cybersecurity, Public-Sector, Insurance. Here, my work has been delivering bleeding-edge technology into effective, comprehensible products for a wide range of contracts.

- Managed 3 concurrent project lines: [Clearwing](#) (Agentic SOC + Pentesting), [AIE](#) (Document Extraction), ATLS (Public Sector)
- Managed 4-person team, translating bleeding edge research into product tech stacks
- Built model-agnostic Agentic Systems across 4 projects, replacing legacy document extraction - LangGraph, [RAG](#), OCR
- Extended document-parsing capability to 1,000+ pages at 1/10 per-page cost - LangGraph, [RLM](#)
- Migrated clients from cloud-hosted SaaS solutions to sovereignly-owned, on-prem solutions - LangGraph, [vLLM](#)
- Synthesized and cataloged training data for the ReAligned/UnCut series of Lazarus Models [Dagster](#)
- Built LLM observability stack to run fully on-prem under HIPAA and SOC 2 constraints - [Langfuse](#)
- Drove evaluation-driven engineering culture, improving document extraction correctness from 20% to 85% - [Langfuse](#)
- Fine-tuned and deployed open-weight models on self-hosted inference infrastructure - [vLLM](#), [Axolotl](#), [LLaMa-Factory](#)
- Built features for Clearwing, replicating 80% of project [Glasswing's findings](#) with open-weight models - Python
- Owned data collection and evaluation for [ReAligned](#), finetunes eliminating propaganda; covered in [The Economist](#)
- Owned production ML platform: k3s cluster, ArgoCD, Dagster pipelines - [Terraform](#), [Kubernetes](#), MLOps

Warmer Founding AI Engineer November 2024 - October 2025

Building the future of Financial Advisory. In this role, I launched two AI products from zero to production: a financial advisor assistant and intelligent client-advisor matching system, optimizing Cost-per-Lead.

- Zero-to-One: founded and scaled AI product from zero to first paying customer - LangGraph, OpenAI, Python
- Built conversational, lead-gen agent, the foundation of the new business model - LangGraph, [LLM](#)
- Achieved <1% hallucinations on golden-set evals through grounding and post-inference validation - LangGraph, [LangSmith](#)
- Architected multi-agent system managing complete client lifecycle, processing 1000+ meetings per day - LangGraph
- Introduced experiment-driven development culture with A/B testing and feature flags - [PostHog](#)
- Built sub-12 second transcript analysis pipeline for real-time processing - [Celery](#), Amazon SQS
- Designed complete agent architecture including RAG, memory, and tool usage systems - [pgvector](#), [RAG](#), LangGraph
- Implemented hybrid-search RAG system across 1,000,000+ client communications - [GraphRAG/LightRAG](#), [pgvector](#)
- Modified open-source dynamic Knowledge Graph, LightRAG, to support [pgvector](#)/Postgres backend - [pgvector](#), Python

AI Consulting

AI Consultant and Contractor

September 2023 - November 2024

Contracted with several early-stage startups, building out AI products.

WorkMate Labs: Workmate is your AI teammate

- Designed and built a Digital Assistant, automating task extraction and email drafting - LLM, TypeScript
- Designed and deployed LLM Prompts for contextual, personal email drafting - ChatGPT, Gemini, LLM
- Designed and deployed custom evaluation framework for measuring LLM performance - LangSmith, TypeScript
- Designed retrieval system for context-aware search of similar sent emails - Retrieval, TypeScript

Stealth Startup: FoodTech, translating conversations into API calls

- Fine-tuned Dolphin Mixtral to improve conversational style - Mixtral, HuggingFace, Python
- Designed and built question-answering workflows inspired by Graph-of-thoughts - OpenAI API, Python
- Designed and built SMS-based chat interface to LLMs - Amazon Bedrock, LangChain, Twilio, Python
- Automated MLOps for LLM-based projects - finetuning, testing, deployment, and performance evaluation
- Designed and deployed IaC for all services - Terraform

Stealth Startup: Generative AI Customers as a service

- Diagnosed and solved a mission-critical problem for scheduling flow of traffic to clients - Math!
- Designed and built API to develop, simulate, and actually orchestrate system traffic flows - Python
- Standardized MLOps approach for use with Stable Diffusion and LLMs - Terraform
- Designed and built tooling to monitor, compare, and research LLM performance - Python, LLM, LLaMa

Owl.co

Software Engineer

January 2023 - September 2023

Returning to the startup world, I designed and built products, catering to the Insurance Industry, with a focus on integrating Machine Learning with Human tasks. Here my focus was on ML products instead of Data scale.

- Led an interdisciplinary team building ML products, automating tasks performed by human investigators: scraping the web, classifying documents, designing ETLs - Clojure, PyTorch, Presto, Spark
- Architected systems to integrate ML inference with human-driven tasks - Clojure, Amazon SageMaker
- Automated hundreds of daily insurance claims investigations with the use of ML, reducing manual investigations by 70%
- Designed and built ETL workflows for analytic databases - Airflow, Spark
- Built dashboards tracking performance of ML models against their human counterparts - Presto, Airflow
- Deployed and maintained infrastructure - AWS CloudFormation, Amazon SageMaker
- Mentored mid-level Engineers on Data Engineering

Sabbatical

October 2021 – January 2023

Oracle Data Cloud, MOAT

Software Engineering Manager

February 2017 - October 2021

I led a highly technical team to create a vast, event-level data store, used as the source-of-truth for the suite of MOAT products. The real-time system processes 1.2M+ records/second, and requires zero downtime. Consequently, I grew a team with high technical aptitude, and emphasized ownership as a core principle in Software Development. Acquired by Oracle.

Engineering Manager, post-acquisition

- Managed and grew team of 7 Data Engineers, ranging from College Recruit to Senior Engineer
- Designed, built, and maintained infra end-to-end: web ingest, stream-processing, hourly ETL to Parquet
- Built multiyear Software Roadmap with Engineering Managers and Product Owners
- Designed and built the event-level data lake, unifying Research, Product, and Engineering datasets - Kafka, Airflow, Parquet
- Processed and stored over 1.2M+ events/second in realtime - Kafka, Golang, Python
- Migrated legacy core business logic to modern systems - Kafka, Airflow
- Managed a team owning 1,000+ instances with a budget of \$340,000+ per month - AWS
- Collaborated with ML and Engineering to design a pipeline powering 5 distinct teams
- Mentored and promoted every Software Engineer on my team

Technical Lead, pre-acquisition

- Managed weekly software releases for core business logic, contributed to by 4 distinct teams
- Designed and built stream-processing applications processing 1.2M+ events/second - Golang, Python, Kafka
- Designed and built system-wide wire protocol - Protobuf
- Built custom software that reduced instance count by 50%, saving over \$2M dollars - Golang
- Built and maintained software end-to-end over 1,000+ AWS instances (c5.xl, r5.8xl)
- Designed "cold storage" data schema - Parquet
- Maintained historical databases, importing 800,000,000+ rows per day - Highly modified Postgres

Prior Research

Harvard IQSS, NECSI

2005 – 2013